

Experimental apparatus for universal spin-motion control in multi-ion registers



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1

Towards Hybrid System Control and Quantum Error Mitigation

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Keep in mind: Want to make clear that this is future work, and not specifically QEM on CV!

Having established and characterised the experimental apparatus and demonstrated its application to *k-copy* randomised benchmarking, this chapter considers future research goals utilising the multi-qubit arrays enabled through the *NPL* trap (section ??), the single addressing system (section ??), and the developed control over both the discrete qubits, as well as the continuous motional modes.

In section 1.1 we establish a set of unitary operations required for universal

continuous-variable (CV) control of the ion motion and discuss the extensions to hybrid spin-oscillator systems. Following on from recent progress in spin-conditioned bosonic operations demonstrated within our group [[bazavan_thesis](#), [saner_thesis](#), [shinbrough_applications_2026](#), 1, 2], we are interested in developing the required addressing capabilities to scale to multiple qubits and motional modes. We outline the method we use to generate such spin-oscillator interactions experimentally, and demonstrate the generation of a squeezed state on the new apparatus. We then outline the next steps required for demonstrating the remaining single- and two-mode interactions, and the possible experiments this will enable.

The second half of this chapter considers an error mitigation technique applicable to both discrete and continuous systems. Developing this, and associated methods on our trapped ion platform will allow substantial reductions of certain errors, with significantly less resources required than full error correction. We give a short introduction to the purpose of quantum error mitigation, followed by an outline of virtual distillation [3–5] — the specific method considered here. We provide an initial demonstrations of this protocol applied to a discrete-variable only system, and explore the performance under both local depolarising and correlated noise due to a common fluctuating control field (see section ?? for details).

We conclude this chapter with a discussion on the application of virtual distillation to hybrid systems, presenting a scalable method within reach of our apparatus.

1.1 Hybrid Spin-Oscillator Control in Trapped Ions

CV quantum computation offers an alternative strategy for encoding quantum information, utilising oscillators, termed *qumodes* [[<empty citation>](#)][Nice to get original ref here](#), as the fundamental quantum object, rather than discrete variable (DV) systems, e.g. the two-level qubit. A single ideal qumode possesses an infinite dimensional Hilbert space, allowing for high (in this case arbitrarily high) density encoding of information in physical carriers. This opens an alternative

avenue for scaling QIPs, requiring dramatically less physical carriers compared to qubit-only systems. The inclusion of both CV and DV subsystems, termed the hybrid model [6], retains the convenient control and readout of discrete qubits while providing the larger state space and native interactions available to qumodes. For example, reference [7] considers a factoring algorithm with the required physical resources independent of the number to be factored. Remarkably, this scheme requires only a single qubit and three oscillators, however requires prohibitively large energy states of the oscillator. Experimentally, we will not be able to realise arbitrarily high energy oscillator states, but it is still an open question of what information density is possible in accessible engineered quantum oscillator.

Hybrid spin-oscillator computation has been studied in the context of superconducting cavity-QED [**<empty citation>**], neutral atom in optical tweezers [**<empty citation>**], and in the trapped-ion platform [**<empty citation>**]. We shall focus on the trapped-ion implementation, where the DV system is the collection of qubits embedded in the electron energy levels, and the qumodes are the shared secular motional modes of the ion crystal within the trapping pseudopotential. Trapped ions benefit from multiple native methods to drive, and couple qubit and qumode: The qumodes may be directly driven through trap potential modulation [**<empty citation>**], or externally applied RF fields [**<empty citation>**]; the qubit may be driven either by optical fields (in the case of optical qubit encoding), or through microwaves (in the case of ground state encoding); and the qubit and qumode may be coupled through qubit-state dependent potential shifts [**<empty citation>**], or through optical fields tuned near to motional-mode frequency sidebands of the qubit-carrier transition [**<empty citation>**]. Taken together, these native control and coupling mechanisms, combined with the long coherence times of both the qubit and qumode, make the trapped-ion platform well-suited for exploring hybrid CV–DV quantum computation.

The apparatus developed in this thesis has demonstrated good qubit and radial motional coherence times, provides access to multiple qubits and qumodes, and includes

a single-addressing system for selectively driving their coupling. These capabilities make it well-placed for studying methods for scaling hybrid-system control.

1.1.1 Theory

Computing on a Hybrid System

For any computing paradigm, we must establish a set of interactions required for universality. In DV-only systems, quantum universality may be realised by a finite set of Clifford operations plus one non-Clifford gate [8]. CV-only universality can analogously be realised by a finite set of Gaussian operations plus one non-Gaussian gate [**<empty citation>**]. The term Gaussian is best understood by the position-momentum representation of the qumode, where the bare oscillator Hamiltonian is given by,

$$H_{\text{osc},0} = \hbar\omega_{\text{osc}}(\hat{x}^2 + \hat{p}^2), \quad (1.1)$$

with quadratures,

$$\hat{x} = \frac{1}{2}(a + a^\dagger), \quad (1.2)$$

$$\hat{p} = \frac{-i}{2}(a - a^\dagger), \quad (1.3)$$

where $a^\dagger$, a are the creation and annihilation ladder operators of the mode with frequency, ω_{osc} . Gaussian operators are then the set of interactions containing at most quadratic terms in $\hat{x}$ and $\hat{p}$, therefore mapping Gaussian input states to Gaussian output states.

The condition for universality of a hybrid CV-DV is somewhat ambiguous as both the CV and DV subsystems are themselves sufficient for universality, however this presents an opportunity: the freedom in selecting the chosen gate-set for universality, combining both Clifford, non-Clifford, Gaussian, non-Gaussian, and spin-oscillator interactions together, allows for selecting efficient, native, and simply accessed interactions of the physical platform. We consider the over-complete instruction set presented in Table 1.1, consisting of spin-conditioned Gaussian and non-Gaussian interactions. The interactions on the bosonic subsystems are themselves universal,

but the spin-conditioning provides a bridge between CV and DV subsystems, allowing for readout and control to be implemented on ancilla qubits. Specifically, we consider a cubic CV instruction set [hillmann_universal_2020, 9, 10], taking the Gaussian operations and supplementing with a tri-squeezing gate, with Hamiltonian,

$$H_T \propto a^3 + a^{\dagger 3}. \quad (1.4)$$

Spin-conditioned tri-squeezing has been shown previously in our group [1], utilising the protocol given in reference [11], here termed the “two-SDF” method. The squeezing, parity, and beam-splitter interactions may be generated by the same method, and have been experimentally demonstrated [1, 12–14]. The displacement operation can be realised either spin-independently by RF fields near resonance to the motional-mode frequencies (section ??), or spin-conditioned using various SDF implementations, such as through a bichromatic field (section ??), through the static Stark-shift gradients due to an optical tweezer [leindecker_state_2026], or by AC Stark-shifts due to counterpropagating Raman beams [empty citation].

Subsection needs summary/transfer sentence.

- Quick overview of the interactions and where we have demonstrated them. Could give flow fields for single-mode operators (could be a nice figure, and I like that view).
- End section by quick discussion on hybrid system, as our generated interactions have spin dependence.

Gate Implementation

- Now how do we realise these interactions? - Present concept of Two-SDF toolkit from TS+RS
- Give Magnus expansion, and commutations (check SS and OB thesis theory sections) **I think this should be a citation and an appendix** - This then gives m , Δ , and Ω (point to table).
- Give short contrast with SB method.

Interaction name	Oscillator H	Gauss.	N_{osc}	Two-SDF method		
				m	$\hat{\sigma}$	Ω
Displacement	$ae^{i\phi} + a^\dagger e^{-i\phi}$	✓	1	-	$\hat{\sigma}_x$	Ω_α
Parity	$aa^\dagger e^{i\phi} + a^\dagger ae^{-i\phi}$	✓	1	+1	$\hat{\sigma}_z$	Ω_2
Beam splitter	$ab^\dagger e^{i\phi} + a^\dagger be^{-i\phi}$	✓	2	+1	$\hat{\sigma}_z$	Ω_2
Squeezing	$a^2 e^{i\phi} + a^{\dagger 2} e^{-i\phi}$	✓	1	-1	$\hat{\sigma}_z$	Ω_2
Trisqueezing	$a^3 e^{i\phi} + a^{\dagger 3} e^{-i\phi}$	✗	1	-2	$\hat{\sigma}_x$	Ω_3

Table 1.1: A suggested set of unitary interactions to compose a universal gate set for continuous variable computation. Using the protocol presented in reference [11], here termed the “Two-SDF” method, we show the required relative detuning setting, m , and the resulting spin conditioning, $\hat{\sigma}$, and coupling strength Ω of the resulting interaction. Ω_α is the typical SDF strength given by equation ??ref eqn in SDF theory sec. The other coupling strengths are found via the 2nd and 3rd terms of the Magnus expansion: $\Omega_2 = 2\Omega_\alpha\Omega_{\alpha'}/\Delta$, $\Omega_3 = 2\Omega_\alpha^2\Omega_{\alpha'}/\Delta^2$.

- Here can also mention SW, cite recent ETH tweezer paper, and that we have even sharper gradient.

To implement the unitaries given in table 1.1, and to retain spin-dependent control of these interactions to fully utilising the large hybrid state space, we use the “two-SDF” method proposed in reference [11]. The non-linear bosonic interactions are effectively driven through applying two linear spin-dependent forces with non-commuting spin conditioning, simultaneously on a single qubit. The two simultaneous SDFs, here labelled SDF_α , and SDF_β , are applied off-resonantly relative to motional modes with creation operators $\hat{a}^\dagger$ and $\hat{b}^\dagger$, respectively. The resulting Hamiltonian is given generally by,

$$\hat{H}_{2\text{SDF}}(t) = \hat{H}_\alpha(t) + \hat{H}_\beta(t) \quad (1.5)$$

$$= \frac{\hbar\Omega_\alpha}{2}\hat{\sigma}_\alpha \left(\hat{a}e^{i(\Delta_\alpha t + \phi_\alpha)} + \hat{a}^\dagger e^{-i(\Delta_\alpha t + \phi_\alpha)} \right) \quad (1.6)$$

$$+ \frac{\hbar\Omega_\beta}{2}\hat{\sigma}_\beta \left(\hat{b}e^{i(\Delta_\beta t + \phi_\beta)} + \hat{b}^\dagger e^{-i(\Delta_\beta t + \phi_\beta)} \right), \quad (1.7)$$

where the SDF strength is Ω_i , the detuning from each mode is Δ_i , the motional phase is ϕ_i , and we ensure the spin-conditioning anticommute $\{\hat{\sigma}_\alpha, \hat{\sigma}_\beta\} =$

0. The unitary dynamics of $\hat{H}_{\text{2SDF}}$ may be evaluated by the Magnus expansion [**magnus_exponential_1954**], with details given in references [11, 13, 14], and in appendix A.8. The Magnus expansion expresses the generator of the unitary evolution as an infinite series of terms, with successive terms proportional to increasingly higher-order nested commutators of the Hamiltonian evaluated at different times. Due to the non-commuting spin conditioning, this series does not truncate, leading to highly non-linear bosonic terms effectively driven via the application of the linear SDFs.

To utilise specific non-linear terms within this infinite series, matching the desired interactions given in table 1.1, we constrain the relative detunings of the two SDFs to $\Delta_\beta = m\Delta_\alpha$, where m is some integer given in the table. This immediately brings components of the Magnus terms matching the desired non-linear $\hat{H}_{\text{osc}}$ into resonance. Furthermore, non-resonant and higher-order terms of the expansion may be arbitrarily suppressed through ensuring $\Delta \gg \Omega_\alpha\Omega_\beta$. In this regime, the unitary dynamics of $\hat{H}_{\text{2SDF}}$ are well approximated by the evolution of time-independent effective Hamiltonians, which for Gaussian operations ($|m| = 1$) are of the form,

$$\hat{H}_{\text{eff,G}} = \frac{\hbar\Omega_\alpha\Omega_\beta}{2\Delta}\epsilon_{\alpha\beta\gamma}\hat{\sigma}_\gamma(iab^{(\dagger)} + h.c.), \quad (1.8)$$

need to add back phases. where $\epsilon_{\alpha\beta\gamma}$ is the Levi-Civita symbols. If $a^\dagger = b^\dagger$, the SDFs are detuned from the same motional mode, leading to squeezing, or parity effective interactions, while acting on two-modes leads to the beam splitter interaction. For the tri-squeezing interaction ($m = -2$) the effective Hamiltonian is,

$$\hat{H}_{\text{eff,T}} = \frac{\hbar\Omega_\alpha^2\Omega_\beta}{4\Delta^2}\hat{\sigma}_\beta(a^3 + a^{\dagger 3}), \quad (1.9)$$

where we have set $a^\dagger = b^\dagger$, as we act on a single mode.

Frequency and spin-conditioning control of two SDFs therefore allow for the generation of the entire proposed hybrid CV-DV gate-set. After a discussion on how we can characterise the oscillator state, we shall present the experimental scheme for generating these interactions, and demonstrate the creation of a squeezed state by this method.

Readout

- We don't have direct access to readout motion.
- So instead we map motional state to qubit.
- We can generally do this via Hadamard test (**generalise to O in figure, and remove conditional I/S**) for phase kickback on spin
- For full state tomography, we can measure characteristic function [**<empty citation>**]**cite Fluhman, OB, SS thesis.**
- Give circuit for tomography using $O = D(\beta)$, and then using σ_x SDFs.
- Define Characteristic function.
- Can then compute Wigner or other representation of choice either through direct Fourier transform, or through ML methods (theorist we spoke to?)

$$\mathcal{P}_a = \frac{1 + \text{Re}(\xi(\beta))}{2}, \quad (1.10)$$

$$\mathcal{P}_a = \frac{1 + \text{Im}(\xi(\beta))}{2}, \quad (1.11)$$

Final state readout of the qumode is done through mapping information to an ancilla qubit, followed by typical state-dependent fluorescence, described in section **??char sec**. The Hadamard test, shown in figure 1.1 (a), offers a general method for measuring the expectation of observable $\hat{O}$ on the oscillator, $\text{Tr}(\hat{O}\rho_{\text{osc}})$, via exclusive measurements of the ancilla qubit. Full state tomography in the form of the characteristic function, $\chi = \text{Tr}(D(\beta)\rho_{\text{osc}})$, may be recovered through conditional displacement operations, i.e. $\hat{O} = e^{i\phi}D(\beta)$, with β defined in equation **??either point to SDF section, or to above gate impl**, and ϕ set to either 0 or $\pi/2$ to measure either $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. As we have access to the native σ_x conditioned spin-dependent force, we may decompose the general Hadamard test to a circuit of native gates, given in figure 1.1 (b), and first described in reference [15], where the proceeding I or $R_y(\pi/2)$ measure $\text{Re}(\chi)$ or $\text{Im}(\chi)$ respectively. The complete description of the oscillator state may now be recovered with arbitrary precision by scanning the complex variable β . We will keep to the characteristic function representation of the state, however we note that other phase space

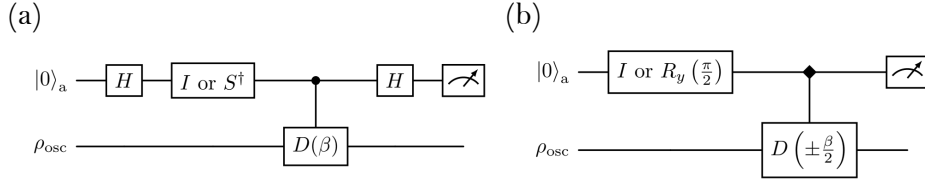


Figure 1.1: State tomography for a single mode through measurement of an ancilla qubit. (a) Displays the Hadamard test method for measuring the characteristic function, $\chi(\beta)$, of the oscillator using a displacement controlled by the qubit. The real and imaginary part of $\chi(\beta)$ is recovered through the application of either the identity, or the $S^\dagger$ gate before the displacement, respectively. (b) An equivalent measurement of $\chi(\beta)$ using native single qubit rotations and spin-dependent forces conditioned on σ_x (represented by the diamond control symbol).

representations, such as the Wigner, and Q-function may be recovered through 2D Fourier transforms of χ [[empty citation](#)][give refs](#).

1.1.2 Experimental Demonstration

Short description of what we will show In our group, we have previously demonstrated higher order squeezing [1], superpositions of non-linear bosonic states [2], beam-splitter [12], and two-mode squeezing interactions. However, due to the constraints of the previous experimental apparatus we were limited to global addressing fields, limiting the application of these hybrid interactions to one- and two-qubits. Here we demonstrate the creation and characterisation of a squeezed state on the newly constructed apparatus, described in chapter ??, utilising the single-qubit addressing system. Single addressing both increases the power efficiency of the interactions, and allows the control to select which qubit is used to couple to the motion — this provides a scalable route to generating hybrid interactions on larger qubit, and qumode registers.

Two Spin Dependent Forces

The individual SDFs are generated by the same bichromatic field method described in section ?? — i.e. simultaneous off-resonant blue- (BSB) and red-sidebands (RSB) of motional mode ω_α are applied. As such, we require four optical fields

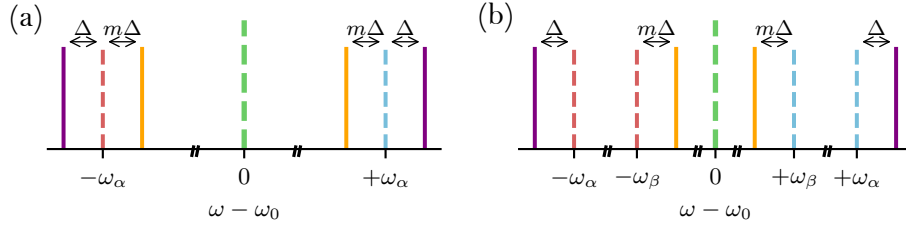


Figure 1.2: The quadchromatic field required for generating non-linear motional interactions via the two-SDF method. (a) For single-mode interactions, both SDF fields are detuned around the sidebands of the same motional mode, ω_α . Integer m is varied to select the desired interaction. (b) For two-mode interactions, each SDF is detuned from a different motional mode. **If want ramp have rid 120238, 119279**

with, frequency, phase, and amplitude control. The four required frequencies of the “quadchromatic” field are shown schematically in figure 1.2 with respect to the carrier frequency, ω_0 . Here, purple lines correspond to SDF_α , and orange lines to SDF_β . In this example we show the SDFs acting on a shared motional mode, but in general they may act on different modes. **Add colour to 2-sdf plot, could also make two examples for BS and sqz?** Similar to the generation of the bichromatic field, we may produce this quadchromatic field through the application of four RF tones onto a single-pass (SP) AOM. These tones are generated using an arbitrary waveform generator (AWG)¹. We cannot control the absolute optical phase at the ion, however we can set relative phases between the four fields by applying phase offsets between the RF tones with respect to some fixed time point. As discussed in section ??, the spin phase of the SDFs is set by the absolute phase of the RSB and BSB, $\hat{\sigma}_\phi = \cos(\phi_s)\hat{\sigma}_x + \sin(\phi_s)\hat{\sigma}_y$. To ensure we saturate the non-commutativity of the spin conditioning of the two SDFs, we require $\pi/2$ phase difference between the two spin phases,

$$\phi_{\alpha,s} - \phi_{\beta,s} = \frac{\phi_{\alpha,\text{RSB}} + \phi_{\alpha,\text{BSB}}}{2} - \frac{\phi_{\beta,\text{RSB}} + \phi_{\beta,\text{BSB}}}{2} = \frac{\pi}{2}. \quad (1.12)$$

In the following demonstration we set $\phi_{\alpha,\text{RSB}} = \phi_{\alpha,\text{BSB}} = \pi/2$, $\phi_{\beta,\text{RSB}} = \phi_{\beta,\text{BSB}} = 0$, however we note that for composing multiple hybrid operations into a circuit, the resulting motional phase of the SDFs can lead to further constraints on the chosen

¹*Sinara Phaser* board

RSB and BSB phases.

The intensity of the quadchromatic field is ramped on and off with a rise envelope of $\sin^2(\pi t/(2t_{\text{ramp}}))$ applied via the AWG, where t_{ramp} is the ramp duration. This ramp suppresses excess off-resonant excitations of the A_1 phase loop term given in the previous section, and so we must select its duration based on the characteristic scale, $\frac{1}{\Delta}$.

We aim to balance the power in each of the 4 tones. This may be calibrated using an optical pick-off before addressing the ion, directed onto a fast photodiode, and measuring the beatnote contrast when applying any pair of tones. In the following squeezing demonstration, we balance the tones comprising the individual SDFs, but do not explicitly balance the power between SDFs, as the desired interaction scales as $\Omega_\alpha\Omega_\beta$, any imbalance leads to non-optimal interaction efficiency, but unlike imbalance of the RSB and BSB, does not increase coherent error terms. We observe that the duty cycle of the SP AOM can dramatically shift the power imbalance between all four tones, and so we take care ensuring that the total RMS power to drive one-, two-, and four-tones is the same as for the idle (no optical pulse) setting of the SP AOM, thus the only transient power drop is that of the ramp sequence.

Squeezing

- Detuning and power selected (only 53 uW!), and predicted strength of interaction?
- Check if ramp is sufficient by looking at σ_z dependence.
- Calibrate probe SDF and get strength.
- Calculate expected squeezing strength.
- Show char function of squeezed state.
- Show r vs duration, with detail of the ramp duration being accounted for.
- Discuss observed strength vs expected strength.
- tickling calibration

We demonstrate the effective squeezing interaction on a single ion and radial motional mode. We are limited to using the radial modes due to the geometry single

addressing system with respect to the trap. We use the upper radial mode with frequency, $\omega_{upper} = 2.87$ MHz, and motional coherence given in section ???. The initial near-vacuum motional state is prepared by Doppler and sideband cooling, with a typical thermal fock state distribution with mean $\bar{n} = 0.03(1)$ **Check for more recent values**, as described in section ???. The spin state is initialised into the ground state by optical pumping, described in section ???. By setting the relative detuning between the SDFs to $m = -1$, and ensuring anti-commuting spin conditioning by setting the relative phases of the quadchromatic field according to equation 1.12, we realise a $\hat{\sigma}_z$ -conditioned squeezing interaction. As the spin-state is prepared in an eigenstate of the spin-conditioned squeezing, we expect squeezing on the qumode, without any change to the qubit state. We use a total quadchromatic beam power of 53 μ W, and set the base detuning, $\Delta = 2\pi$ 50 kHz. We suppress excess excitation of the off-resonant error terms by using a ramp of $t_{\text{ramp}} = 40$ μ s. The state is characterised by the tomography method given in section 1.1.1, with a discretization set to collect a 35×35 grid of displacement values, β , and $N = 300$ repeat samples to reduce shot noise. The full dataset for one of the tomographic scans takes around 1 hour to collect, and so we interleave mode calibration experiments (section ??) to run every 5 minutes.

The on-resonance splitting SDF used for tomography must have its strength, Ω_{SDF} , calibrated to infer the magnitude of β applied. We use a total bichromatic beam power of 40 μ W, and through fitting the $\hat{\sigma}_z$ expectation for the time-dependent splitting of a vacuum state,

$$\langle \hat{\sigma}_z \rangle = e^{-2|\beta|^2}, \quad (1.13)$$

with

$$|\beta| = \frac{\Omega_{\text{split}} t}{2}. \quad (1.14)$$

We find $\Omega_{\text{split}} = 2\pi$ 7.2(4) kHz.

The pulse sequence for squeeze state generation and characterisation is given in figure ??, and the resulting real part of the characteristic function is shown in

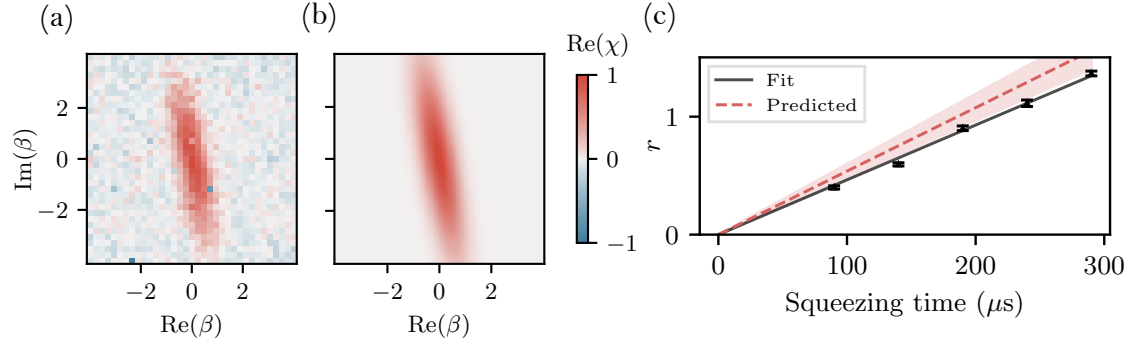


Figure 1.3: Figure: Sqz characteristic function with varying duration

Demonstration of the squeezing interaction through the two-SDF method. (a) The characteristic function, $\text{Re}(\chi(\beta))$, of the squeezed state measured using the state tomography method given by figure 1.1. (b) The simulated state taking fit parameters inferred by equation ?? . (c) The fitted squeezing parameter, r , as a function of two-SDF pulse duration.

figure 1.3 (a). Fitting the data to equation ?? we may extract both the squeezing axis, ϕ , and the strength, r , where r is given by,

$$r = |\Omega_{\text{eff},S}|t = \frac{\Omega_\alpha \Omega_\beta}{2}t. \quad (1.15)$$

An example fit is shown in panel (b), with extracted $r = 0.90(2)$, for a quadchromatic FWHM pulse duration of $t_{\text{sqz}} = 190 \mu\text{s}$. Panel (c) displays the squeezing strength as a function of t_{sqz} , with fit values given by the points, and the black line is a linear fit. Assuming that the two SDF strengths are well balanced, $\Omega_\alpha = \Omega_\beta$, and taking the 53 μW used for the quadchromatic field compared to the SDF strength calibration at 40 μW , we find an expected squeezing strength of $\Omega_{\text{eff},S} = 2\pi \ 0.86(9)$ kHz. This is shown as the red dashed line and shaded region. The fitted strength of $\Omega_{\text{eff},S} = 2\pi \ 0.74(1)$ kHz is lower than the expected, suggesting that either we are not sufficiently balancing the strengths of the two simultaneous SDFs, or that the power stabilisation routine for 4-tones is inaccurate. Further exploration is needed, including careful quantification of the individual SDF strengths constituting the quadchromatic field.

Wrap up section.

1.1.3 Outlook

Hybrid systems offer access to large Hilbert spaces, even with modest ion numbers. In this section we outlined a convenient set of native interactions for universal hybrid qubit-qumode gates. We experimentally demonstrate the squeezing interaction generated via the “two-SDF” method of reference [11]. This demonstration replicated previous work using the “two-SDF” method [1], however now shown on the newly built apparatus utilising a single-ion addressing system, and applied to the radial motional modes.

Enabled by the single-addressing, and improved motional coherence times of the new system, we are particularly interested in exploring short hybrid circuits composed of multiple interactions acting on an array of qubits and qumodes. Specifically, we are interested in utilising the hybrid system with increased robustness against errors through the use of quantum error mitigation methods applicable to both spin and oscillator subsystems. We outline one such method in the next section, with a preliminary demonstration on the spin subsystem.

Standing wave extension

Limitations

Using a single multi-ion chain to access multiple qubits and qumodes for scaling hybrid QC on the trapped-ion platform will be ultimately limited by spectral crowding of the motional modes, i.e. the average frequency spacing between neighbouring modes decreases as the number of accessible modes increases.

The “two-SDF” method requires frequency selectivity to drive bosonic interactions exclusively on the chosen mode. To selectively drive a single mode, the magnitude of the detuning Δ must follow $\Delta \ll \delta\omega$, where $\delta\omega$ is the frequency difference to the nearest neighbouring mode. As discussed in the “two-SDF” method section, the strength of each SDF must be less than the base detuning, $\Omega_\alpha \ll \Delta$, to well approximate the effective non-linear unitary dynamics. As a result of these two requirements, the mode-frequency splitting directly limits the attainable strength

of the non-linear interaction in the adiabatic regime, resulting in increasing gate duration with increasing register size.

A possible mitigation would be to rather scale the hybrid system using a collection of fixed-length chains with adequate mode-frequency splitting, and to transport quantum information between these chains either through qubit transport [**<empty citation>**], optical networks [**<empty citation>**], or through inducing collective interactions between separate ion-chains ??.

Another mitigation strategy could come from the pulse-shaping approach demonstrated for use in fast two-qubit entangling gates in the non-adiabatic regime [16], i.e. when $\Delta > \delta\omega$. Introducing extra degrees of freedom in the pulse amplitude envelope could allow for driving the desired interaction on the selected modes, while returning all spectator modes to their initial state by the end of the pulse. However, we note that the accumulation of non-global geometric phases, and possible non-commutativity of interactions on shared qubit-oscillator states could prohibit the use of such a pulse-shaping scheme. Ultimately, the desired interaction, mode frequency spectra, and target gate duration will determine the dominant coherent errors. We leave such an analysis for future work.

1.2 Virtual Distillation

Short summary of section Here we develop experimental tools for demonstrating VD on the discrete variable portion of the full hybrid system.

1.2.1 Error Mitigation

Noise coupled to a quantum system results in unintended coherent, and incoherent evolution. Deviations of the realised state from the target state are known as errors, and, if left unchecked, lead to the failure of quantum algorithms. This failure can either be due to a reduction of signal to noise ratio, or a bias in the measured expectation value, resulting in misleading conclusions. Ultimately, the goal of quantum error correction, QEC, is to detect and correct these errors

through embedding information in a larger Hilbert space [**qec_review**]. Full QEC will allow for the execution of arbitrarily long algorithms ensuring errors (of up to a certain distance) do not accumulate over time. However, current QEC methods require large overheads in both qubit number and circuit depth, and are therefore not feasible on current quantum hardware. Furthermore, QEC is formulated for digital “gate-based” quantum computers, which although promise to be universal, pay for that privilege in often inefficient decompositions of target unitaries into a discrete set of gates. This poses an opportunity for near term devices to show quantum advantage using either short depth, uncorrected circuits, or analog quantum simulators, which although not universal, can provide meaningful results for specific problems [**nisq_review**]. Particularly, phrasing the problem in terms of expectation values of an observable on a physical final state, rather than a sequence of measurements on a set of logical state, allows for a more efficient use of resources. As a tactile example, consider the problem of finding the ground state energy of a Hamiltonian. This can be done by preparing a trial state, and directly measuring the expectation value of the Hamiltonian on that state, building up statistics over many samples [**variational_qc_review**]. Increasing the number of samples will reduce the statistical error, however, if the trial state is not prepared perfectly due to either environmental noise, or to imperfect application of unitary evolutions, the measured expectation value will be biased. To this end, quantum error mitigation (QEM) techniques have been developed to reduce the bias of measured expectation value of an observable at the expense of requiring more samples. Reference [17] defines QEM as “... algorithmic schemes that reduce the noise-induced bias in the expectation value by postprocessing outputs from an ensemble of circuit runs, using circuits at the same noise level as the original unmitigated circuit or above.” This highlights the crux of how information on the system is collected by a QEM implementation, which I shall explore with the following thought experiment on the “perfect” QEM algorithm. Consider the problem of measuring the expectation value of an observable $\hat{O}$ on the target state ρ_{id} . The state is prepared by a quantum circuit, which due to noise, instead prepares the mixed state $\rho = (1 - \lambda)\rho_{id} + \lambda\rho_{err}$,

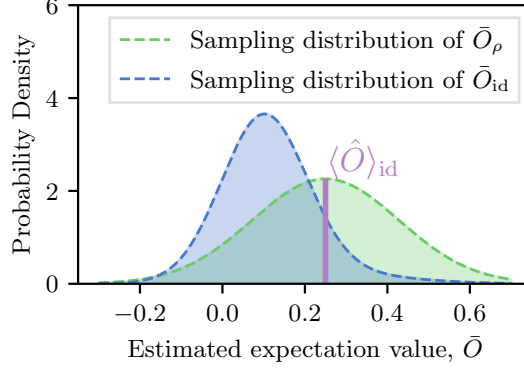


Figure 1.4: Figure: unmitigated vs mitigated expectation value, showing how bias is reduced at the expense of more samples. $\rho = 0.1\rho_{id} + 0.9\rho_{err}$. Blue curve is PDF for N shots, green is PDF for ideal shots, with $N_{id} = 0.1N$.

A schematic explanation of the goal of quantum error mitigation. Deviations of the measured expectation value of an observable $\hat{O}$ on a noisy state, ρ , and on the ideal state ρ_{id} may be separated into components originating from shot noise, and from an underlying bias. Shot noise sets the variance of the distribution of estimated expectation values, while the underlying bias shifts the center of the distribution. Error mitigation reduces bias at the cost of increasing variance. Here we demonstrate this by the ideal QEM method introduced in section 1.2.1. The underlying noisy state is given by $\rho = (1 - p)\rho_{id} + p\rho_{err}$. N shots are taken measuring $\text{Tr}(\hat{O}\rho)$. The ideal QEM flags and discards shots originating from ρ_{err} , leaving $(1 - p)N$ shots of the ideal state to estimate $\langle \hat{O} \rangle$.

where ρ_{err} is the error state. Each measurement of the observable $\hat{O}$ on the noisy state ρ will return a value o_i , and the average of these measurements will converge to the expectation value $\langle \hat{O} \rangle = \text{Tr}[\hat{O}\rho]$. Given S total samples, the final result will be,

$$\langle \hat{O} \rangle = \frac{1}{S} \left(\sum_{i=1}^{S_{id}} o_i + \sum_{j=1}^{S_{err}} o_j \right),$$

where S_{id} and S_{err} are the number of samples that return a measurement outcome from the ideal and error states respectively, with $S = S_{id} + S_{err}$. The bias in the measured expectation value from the ideal is then given by contributions from the samples S_{err} . An ideal QEM algorithm would flag these error samples, and remove them from the final average, resulting in a bias free expectation value using the restricted samples S_{id} .

- Error mitigation motivation. Mainly on contrasting to quantum error correction.

- QEM is framed as finding expectation value of observable, rather than maintaining coherence for a long computation.
- How QEM reduces bias (at the expense of required samples). Look at Cai paper for structured argument.

1.2.2 Virtual Distillation Protocol

Purification

Estimate expectation value of observable on a state without ever physically preparing the state. Purification here refers to approaching the dominant pure state of a mixed state. The k th degree purified state is given by,

$$\rho_{\text{corrected}}^{(k)} = \frac{\rho^k}{\text{Tr}[\rho^k]} = \sum_{i=1}^d \frac{\lambda_i}{\sum_{i=1}^d \lambda_i^k} |\psi_i\rangle \langle \psi_i|, \quad (1.16)$$

where d is the dimension of the Hilbert space, λ_i are the eigenvalues of ρ , and $|\psi_i\rangle$ are the corresponding eigenvectors. If we assume that one eigenvalue is dominant, $\lambda_1 \gg \lambda_{i>1}$, then the normalised purified state exponentially approaches the dominant eigenvector, $\rho_{\text{corrected}}^{(k)} \rightarrow |\psi_1\rangle \langle \psi_1|$, as k increases. The hope, which we shall verify in section XX on coherent mismatch, is that the dominant eigenvector of the noisy state closely approximates the ideal state, and therefore effective measurements on the purified state yield accurate expectation values of the ideal state. The estimated expectation value of measuring an observable, $\hat{O}$, on this purified state is then,

$$\langle \hat{O} \rangle_{\text{corrected}} = \text{Tr}[\hat{O} \rho_{\text{corrected}}^{(k)}] = \frac{\text{Tr}[\hat{O} \rho^k]}{\text{Tr}[\rho^k]}. \quad (1.17)$$

Next, how to access this corrected state on which to make these measurements? Kozcor and Huggins show that this can be accessed without explicit preparation of the purified state. Instead the numerator and denominator may be found through preparing k copies of the noisy state, applying some derangement operation to entangle all copies together, and measuring the desired observable on one of the copies. This can be understood through the following identity,

$$\text{Tr}[\hat{O} \rho^k] = \text{Tr}[\hat{O}^i S^{(k)} \rho^{\otimes k}], \quad (1.18)$$

where $S^{(k)}$ is the derangement operator, which permutes the k copies of the state, and $\hat{O}^i$ is the observable acting on the i th copy. We shall define the derangement operator explicitly as the cyclic permutation of k objects,

$$S^{(k)} |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_k\rangle = |\psi_2\rangle \otimes |\psi_3\rangle \otimes \dots \otimes |\psi_k\rangle \otimes |\psi_1\rangle. \quad (1.19)$$

Huggins provides a proof of this identity, which I reproduce here for the simple example of $k = 2$. The derangement operator for two copies is simply the swap operator, $S^{(2)} = \text{SWAP}$, and we will assume the observable acts only on the first copy, giving $\hat{O}^1 = \hat{O} \otimes \mathbb{I}$. The identity is then,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (1.20)$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \langle i_1 | \hat{O} | k_1 \rangle \delta_{i_2, k_2} \langle k_1, k_2 | S^{(2)} | j_1, j_2 \rangle \langle j_1 | \rho | i_1 \rangle \langle j_2 | \rho | i_2 \rangle \quad (1.21)$$

$$= \sum_{\substack{i_1, i_2 \\ j_1, j_2 \\ k_1, k_2}} \hat{O}_{i_1, k_1} \delta_{i_2, k_2} \delta_{k_1, j_2} \delta_{k_2, j_1} \rho_{j_1, i_1} \rho_{j_2, i_2} \quad (1.22)$$

and contracting the Kronecker deltas, we find,

$$\text{Tr}[\hat{O}^1 S^{(2)} \rho^{\otimes 2}] \quad (1.23)$$

$$= \sum_{i_1, j_1, j_1} \hat{O}_{i_1, j_2} \rho_{j_1, i_1} \rho_{j_2, j_1} \quad (1.24)$$

$$= \text{Tr}[\hat{O} \rho^2], \quad (1.25)$$

with the last equality following from the definition of matrix multiplication.

The denominator of the corrected expectation value is found by replacing the observable with the identity and measuring the expectation value of the derangement operator on k copies of the noisy state.

Access to this corrected expectation value is then possible if we can measure the multi-qubit observable $\hat{O}^i S^{(k)}$ on k copies of the noisy state. We will discuss two methods of implementing this measurement. First, a Hadamard test, given in Fig. ??, which uses an ancilla qubit to control the derangement and desired observable operation, and then subsequent measurement of the ancilla only. The second method, given in Fig. ??, uses a B-matrix to implement the derangement

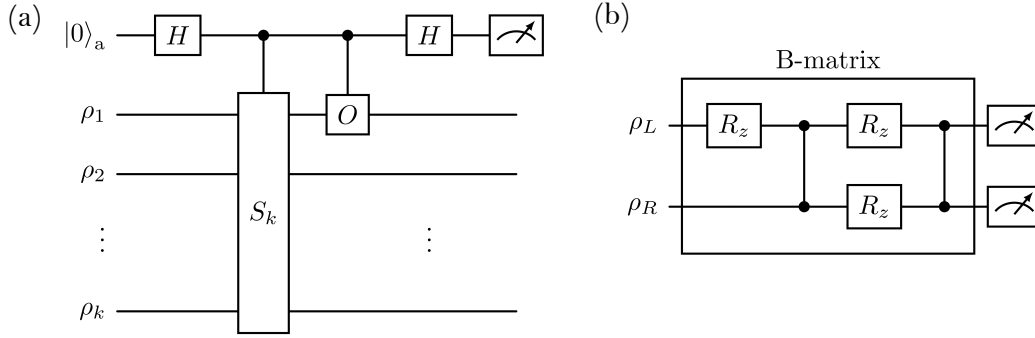


Figure 1.5: Figure: circuit diagram for ancilla based virtual distillation, Hadamard test, derangement operator by CSwap Two circuits for implementing virtual distillation. (a) Hadamard test method using controlled derangement, S_k , over k copies, and controlled observable, $\hat{O}$, applied on one copy. The ancilla qubit is measured to extract $\text{Tr}(\hat{O}\rho^k)$. Repeating the circuit without the controlled observable measures normalisation factor, $\text{Tr}(\rho^k)$, used to recover the virtually purified result, $\langle \hat{O} \rangle_{\text{VD}}$. (b) Two-copy implementation using a B -circuit, here decomposed into two entangling gates and single qubit phase gates. Measurement of both qubits in the computational basis, and classical postprocessing recovers $\text{Tr}(Z\rho^2)$ and purity, $\text{Tr}(\rho^2)$, used to recover $\langle Z \rangle_{\text{VD}}$.

operation, and then measures the observable on one of the copies. The Hadamard test is more general, as it can be used to measure any observable, however, it requires an additional ancilla qubit and controlled operations. The B-matrix method has better qubit number and operation depth efficiency, however, requires simultaneous diagonalisation of $S^{(k)}$ and $\hat{O}^i S^{(k)}$, which is not always possible for an arbitrary observable and derangement operator.

Next, need to derive scaling of bias with copy number. And finally, discuss the variance of measurement with shot number (and how this scales with copy number?).

Circuit

Ancilla method

B-matrix protocol

If we can diagonalise both S and SO . Can apply transversally

On hybrid systems

It can be used for both spin and bosonic objects

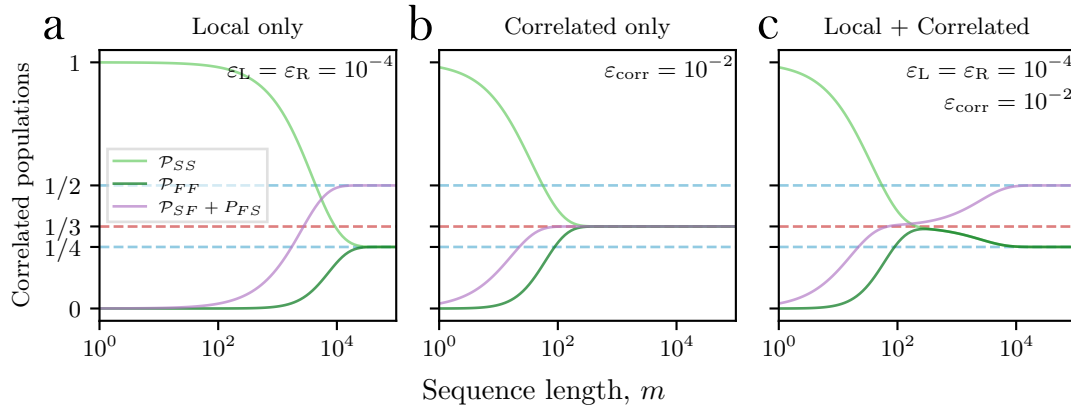


Figure 1.6: Figure: pi-4 MS gate calibration. Figure: Heuristic signal of B-matrix working. Tomography would be best!

1.2.3 Experimental Demonstration

Short description of following results.

Implementation

Protocol, circuit diagram, hardware involved.

B-matrix Calibration

pi-4 gates, single addressing

Injecting Spatial Correlations

two copy, B matrix demonstration

Analysis of this data.

1.3 Proposed experiments and technical requirements

How CVDV and VD combine.

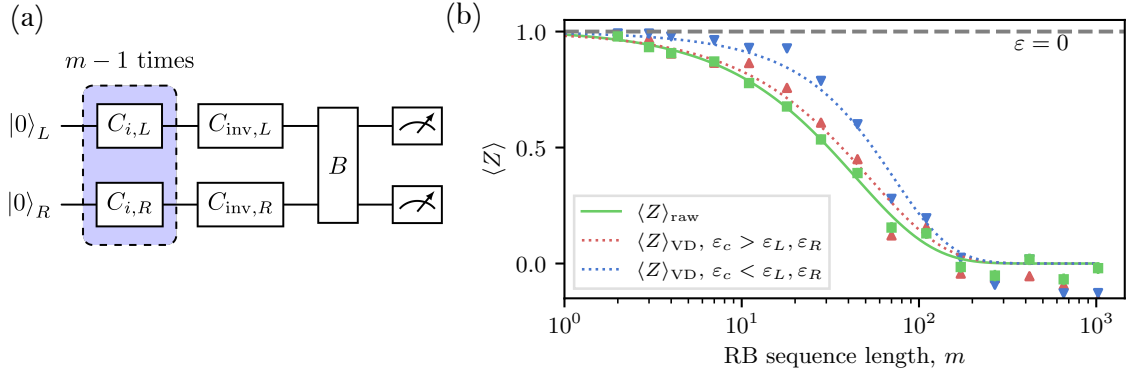


Figure 1.7: Figure: Circuit diagram of RB + VD, exp data for noise floor for two copy B-matrix, with and without spatial correlations

Verifying the performance of virtual distillation under either dominant local depolarising or correlated errors. Structured errors were injected through a fixed frequency detuning of the 729-nm laser, and the application of a sequence of random Clifford gates. In the case that identical Clifford gates, $C_{i,L} = C_{i,R}$, are applied on both qubits, the detuning error is twirled to a *common noise* channel, while in the case that Clifford gates are sampled independently for both copies, the detuning is twirled to local depolarisation channels. The local and correlated injected error-per-Clifford for both regimes are quantified using k -copy RB. Panel (a) shows the circuit of injected errors followed by the B -circuit. Panel (b) shows the effective reduction in error-per-Clifford by applying the virtual distillation circuit. The green points correspond to the raw single-qubit decay due to the physical errors, the solid green line is a fit of this decay to measure a total error-per-Clifford. The red and blue points correspond to the error mitigated Z observables for the dominant correlated errors and local depolarising errors respectively. The dotted lines correspond to the expected error mitigated single qubit decays inferred by the model given in equation ??, and the error-per-Cliffords measured by k -copy RB.

Extending to > 2 copies

plan for c-swap implementation.

What is difficult, what numbers we need (gate fidelities, shot number)

Virtual Distillation on Hybrid Systems

Cite Quantum virtual cooling

Why important to develop

What errors does this mitigate?

How it looks on a thermal state and on various quantum states (sqz, coherent, fock)

What is difficult, what numbers we need (gate fidelities, shot number)

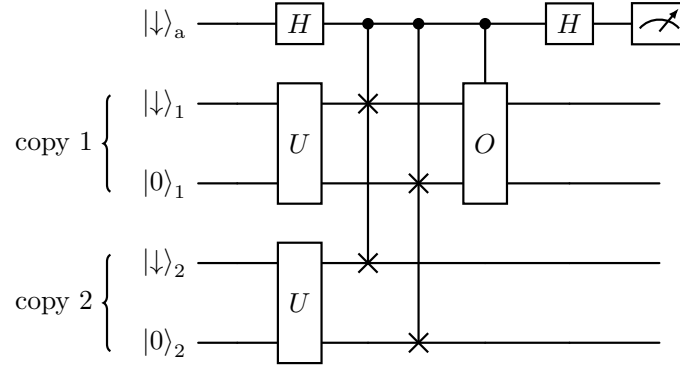


Figure 1.8: Figure: Fredkin gate on oscillator, and how to compose derangement transversally. Proposed circuit for applying virtual distillation to a hybrid system. Using the ancilla method for measuring $\text{Tr}(\hat{O}\rho^k)$, we may decompose the controlled derangement to the transversal application of Fredkin gates.

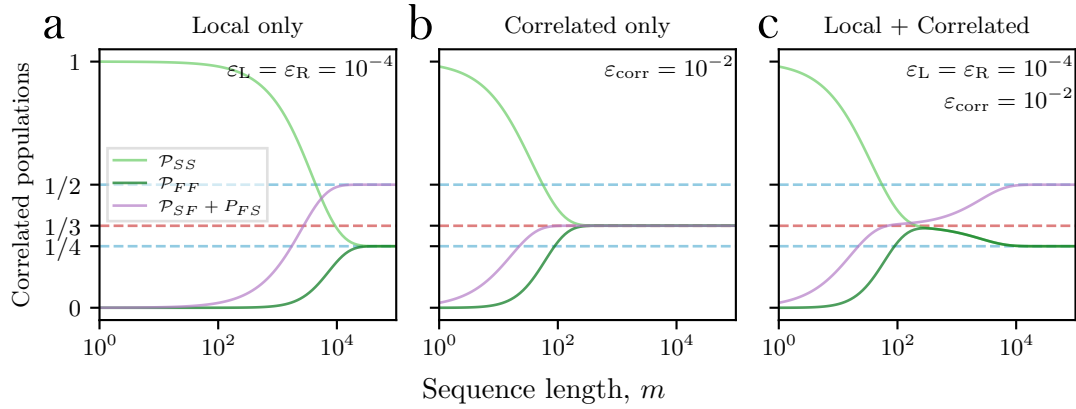


Figure 1.9: Figure: Simulation of tri-sqz state with VD to show negativity!

1.4 Outlook

Broader perspective of these QEM research directions, and also how we can play with our larger CVDV system Can generate sqz using SW

Appendices

A

Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

A.5 Gate Fidelity

We derive the fidelity of a noisy gate implementation with respect to the ideal unitary.

A quantum channel,

$$\mathcal{E} : \rho \mapsto \rho, \tag{A.1}$$

which executes ideal unitary, U , with some error, has an average operation fidelity given by,

$$F(\mathcal{E}, U) = \int d\psi \langle \psi | U^\dagger \mathcal{E}(|\psi\rangle \langle \psi|) U | \psi \rangle, \tag{A.2}$$

where the integral is taken over the uniform Haar measure on the space of pure states [18, 19]. We shall see that the language used for this analysis of single qubit

gate fidelities closely resembles the group theory approach describing randomised benchmarking. The Haar measure is relevant here to consider the fidelity of the operation independent of the initial state (assuming it is pure), which is exactly the enforced criteria through Clifford twirling during RB. For a single qubit, this can be expressed as an integral over the Bloch sphere. If we further assume that the error channel is unitary, as is the case of a detuned channel considered here, we shall find that the fidelity follows

$$F(V, U) = \frac{1}{d(d+1)} \left(|\text{Tr}(U^\dagger V)|^2 + d \right), \quad (\text{A.3})$$

where V is the error unitary, and d is the dimension of the Hilbert space.

Here we shall give a derivation of this result, by considering the symmetry imposed by the Haar measure. The standard fidelity of two unitaries following from equation A.2 is given by

$$\begin{aligned} F(V, U) &= \int d\psi \langle \psi | U^\dagger V | \psi \rangle \langle \psi | V^\dagger U | \psi \rangle, \\ &= \int d\psi |\langle \psi | W | \psi \rangle|^2, \end{aligned} \quad (\text{A.4})$$

where $W = U^\dagger V$. Using the identity $\text{Tr}(A)\text{Tr}(B) = \text{Tr}(A \otimes B)$, we can rewrite the fidelity as a two-copy expectation value,

$$F(V, U) = \text{Tr} \left((W \otimes W^\dagger) \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2} \right), \quad (\text{A.5})$$

where we have now separated the fidelity into a part dependent on the error channel, and a part equivalent to the second moment of the Haar measure, explicitly rewritten as

$$\mathcal{M}_2 = \int d\psi (|\psi\rangle \langle \psi|)^{\otimes 2}. \quad (\text{A.6})$$

We now consider two apparent symmetries of $\mathcal{M}_2$: first, the Haar measure is invariant under unitary transformations, and so our expression must commute with $U \otimes U$ for any unitary U ; second, the two states in the tensor product are identical, and so $\mathcal{M}_2$ must be invariant under the swap operator S which exchanges the two copies. These

two symmetries are sufficient to show that $\mathcal{M}_2$ is proportional to the projector onto the symmetric subspace of the two-copy Hilbert space, which is given by

$$\Pi_{\text{sym}} = \frac{1}{2}(\mathbb{I} + S). \quad (\text{A.7})$$

To normalise the moment (as the Haar measure is a probability distribution), we enforce $\text{Tr}(\mathcal{M}_2) = 1$. In this two-copy expression, it is apparant that $\text{Tr}(I) = d^2$ and $\text{Tr}(S) = d$, resulting in,

$$\mathcal{M}_2 = \frac{1}{d(d+1)} (\mathbb{I} + S). \quad (\text{A.8})$$

Incorporating this result into the fidelity expression, we find

$$F(V, U) = \frac{1}{d(d+1)} \text{Tr} \left[(W \otimes W^\dagger)(\mathbb{I} + S) \right], \quad (\text{A.9})$$

where we can evaluate the two terms using the identities $\text{Tr}(A \otimes B) = \text{Tr}(A)\text{Tr}(B)$ and $\text{Tr}((A \otimes B)S) = \text{Tr}(AB)$, the latter of which is highly relevant to the mechanism of virtual distillation, described in chapter ???. This results in the final expression for the fidelity of a unitary error channel with respect to a target unitary, given above by equation A.3.

A.6 Projectors of $Cl_1^{\otimes 2}$

We outline the method of recovering the α parameters described in section ?? using projectors onto the irrep spaces. A special thanks to Iver Øvergaard who performed this analysis applied to the *common noise* channel and gave permission for reproducing the results here **IRO please check you are happy with me to include this appendix.**

The decay due to a single rotation acting identically on both qubits which, in the Pauli-transfer matrix (PTM) representation, can be written as $\Lambda^{\mathcal{P}} = \text{diag}(1, R, R, R \otimes R)$, where superscript $\mathcal{P}$ represents the block diagonal matrix is ordered by the two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z\}$, and

$R = R(\vec{n}, \theta) \in SO(3)$ describes a rotation by θ around $\vec{n}$. Finding the vector $\vec{\alpha}$ can now be done using Schur's lemma for the case of no mixing between irrep spaces:

$$\tilde{\Lambda}^{\mathcal{P}} = \sum_{\lambda} \alpha_{\lambda} \Pi_{\lambda}, \quad \alpha_{\lambda} = \frac{1}{d_{\lambda}} \text{Tr}(\Pi_{\lambda} \Lambda^{\mathcal{P}}). \quad (\text{A.10})$$

Here, Π_{λ} denotes the PTM projector onto the irrep space labelled λ and with dimension d_{λ} . It follows immediately that the identity sector is fixed, while the left- and right-qubit sectors produce $\alpha_L = \alpha_R = \text{Tr}(R)/3$. In principle, α_{λ} for the two-qubit sector can be evaluated by explicitly constructing the projectors from the basis operators in Table ???. However, in this case the symmetries of these irrep spaces allow the projectors to be expressed more simply in terms of operators whose traces are easily evaluated:

$$\Pi_{tr} = T, \quad (\text{A.11})$$

$$\Pi_{dt} = D - T, \quad (\text{A.12})$$

$$\Pi_{so} = \frac{1}{2}(I_9 + S) - D, \quad (\text{A.13})$$

$$\Pi_{as} = \frac{1}{2}(I_9 - S), \quad (\text{A.14})$$

where $T = 1/3 \sum_{ij} |ii\rangle \langle jj|$, $D = \sum_i |ii\rangle \langle ii|$, and $S = \sum_{ij} |ji\rangle \langle ij|$ are the trace projector, diagonal projector, and swap operator, respectively. The required traces are

$$\text{Tr}(I_3 R) = \text{Tr}(R) = (1 + 2c), \quad (\text{A.15})$$

$$\text{Tr}(I_9(R \otimes R)) = (\text{Tr}(R))^2 = (1 + 2c)^2, \quad (\text{A.16})$$

$$\text{Tr}(T(R \otimes R)) = \frac{1}{3} \text{Tr}(R^T R) = 1, \quad (\text{A.17})$$

$$\text{Tr}(D(R \otimes R)) = \sum_i R_{ii}^2 = c^2 + 2c + (1 - c)^2 \kappa, \quad (\text{A.18})$$

$$\text{Tr}(S(R \otimes R)) = \text{Tr}(R^2) = 4c^2 - 1, \quad (\text{A.19})$$

where $c = \cos(\theta)$, and $\kappa = n_x^4 + n_y^4 + n_z^4$ is an anisotropy parameter reflecting the fact that Clifford twirling enforces symmetry under permutations of the Pauli axes, rather than under arbitrary rotations. Substituting these traces into equation A.10

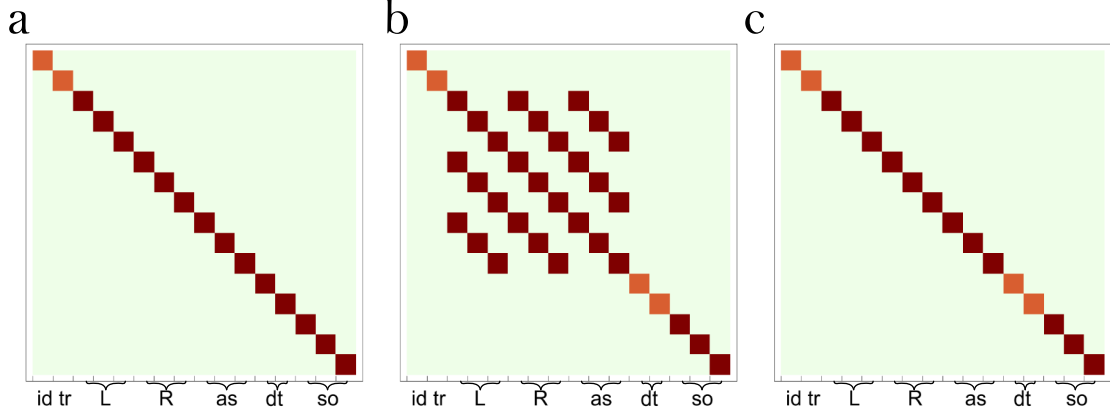


Figure A.1: Liouville operator representation of error channels after twirling over $Cl_1^{\otimes 2}$, ordered by the irrep basis (Schur basis). (a) *Common noise*, (b) *Continuous ZZ*, (c) *Stochastic ZZ*. Orange blocks represent an entry of 1, green represent 0, and red represent a decay with dependence on m .

and using equation A.11 yields

$$\alpha_L = \alpha_R = \alpha_{as} = \frac{1}{3}(1 + 2c), \quad (\text{A.20})$$

$$\alpha_{tr} = 1, \quad (\text{A.21})$$

$$\alpha_{dt} = \frac{1}{2}(c^2 + 2c - 1 + (1 - c)^2\kappa), \quad (\text{A.22})$$

$$\alpha_{so} = \frac{1}{3}(3c^2 - (1 - c)^2\kappa). \quad (\text{A.23})$$

A.7 ZZ-type noise

In the main text we explore a separable error channel that induces correlations between qubits in a register. Here we shall explore entangling type noise, which although not prevalent on the ion-trap qubit platform, is a dominant error source in superconducting qubits.

The always on ZZ-type error common to superconducting qubits is described by the Hamiltonian,

$$H_{ZZ} = \frac{J_{ZZ}}{2} Z \otimes Z, \quad (\text{A.24})$$

where J_{ZZ} is the coupling strength. Generally, H_{ZZ} will couple the subrepresentations of the 3-irrep, given in table ???. We can show the qualitative difference

in the subspace dynamics by plotting the Liouville representation of the twirled *common noise*, and the *ZZ*-type noise. Specifically we plot $\tilde{\Lambda}^m$, where m is the sequence length. Figure ?? plots the maps in the Schur basis ordered so isotypic representations are adjacent to one another. Orange blocks have entries of 1, light green blocks are 0, and red blocks represent some decay parameters dependent on m . Panel (a) shows the fully diagonal *common noise* transfer matrix, where we can see the invariant trace subspace that was noted in the main text. Panel (b) shows the general case for twirled *ZZ*-type noise, given by equation A.24. The off-diagonal entries exactly couple the three subspaces of the 3-irrep. We note that the resulting block diagonal form is inconvenient to express as a general multi-exponential decay profile as given in the main text for *common* and local depolarising type noise. However, the resulting block diagonal Liouville form may be used directly with experimental data to fit these parameters of interest. Further, we may still explore the expected asymptotic decays due to this model, and find that under dominant *ZZ*-type noise, we expect an asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 1/2$.

We may recover a simple diagonal Liouville form in the case of stochastic *ZZ* noise, given by,

$$\mathcal{E}_{ZZ}(\rho) = (1 - p_{ZZ})\rho + p_{ZZ}(Z \otimes Z)\rho(Z \otimes Z), \quad (\text{A.25})$$

where p_{ZZ} is the probability of a correlated phase flip error on both qubits. In the irrep basis, the following sector decays are found for the stochastic *ZZ* error,

$$\begin{aligned} \alpha_L = \alpha_R = \alpha_{so} = \alpha_{as} &= 1 - 4p_{ZZ}/3, \\ \alpha_{tr} = \alpha_{dt} &= 1. \end{aligned} \quad (\text{A.26})$$

We show this dependence in figure A.1 (c). The same intermediate asymptote of $\mathcal{P}_{SS}|_{m \rightarrow \infty} = 0.5$ is expected.

The difference in correlated decay structure due to *common*, depolarising, and *ZZ*-type errors suggests that twirling over $Cl_1^{\otimes 2}$ both simplifies a generic error map to be measurable via RB, but also, retains some noise-dependent structure

of the correlated decay profiles. Designing optimal initial and final states, as well as optimal sampling strategies for extracting a larger set of error parameters covering multiple physical error mechanisms is left for future work. A possible strategy to avoid introducing arbitrary error channels, is to be agnostic of the physical error mechanisms, and rather directly extract the $\vec{\alpha}$ and $\vec{m}$ parameters described in section ???. Analogous protocols to this is described by character RB [20], and tomography via RB [21].

A.8 Magnus expansion

The unitary evolution due to a time dependent Hamiltonian may be written in the ordered, short-time limit as,

$$U(t) = \lim_{N \rightarrow \infty} e^{\frac{-i}{\hbar} \hat{H}(t_N) \Delta t} \dots e^{\frac{-i}{\hbar} \hat{H}(t_2) \Delta t} e^{\frac{-i}{\hbar} \hat{H}(t_1) \Delta t} = e^{A(t)}, \quad (\text{A.27})$$

where $\Delta t = t/N$, and $t_n = n\Delta t$, and $A(t)$ is a time dependent exponent, which is expanded in the Magnus construction to,

$$A(t) = \sum_{k=1}^{\infty} A_k(t). \quad (\text{A.28})$$

If the Hamiltonian at different times does not commute, then the first three $A_k(t)$ terms are given by,

$$A_1(t) = \frac{-i}{\hbar} \int_0^t dt_1 \hat{H}(t_1) \quad (\text{A.29})$$

$$A_2(t) = -\frac{1}{2\hbar^2} \int_0^t \int_0^{t_1} dt_1 dt_2 [\hat{H}(t_1), \hat{H}(t_2)] \quad (\text{A.30})$$

$$A_3(t) = \frac{i}{6\hbar^3} \int_0^t \int_0^{t_1} \int_0^{t_2} dt_1 dt_2 dt_3 \left([\hat{H}(t_1), [\hat{H}(t_2), \hat{H}(t_3)]] + [\hat{H}(t_3), [\hat{H}(t_2), \hat{H}(t_1)]] \right). \quad (\text{A.31})$$

Given the “two-SDF” Hamiltonian in equation 1.5, we can evaluate $A(t)$ to a chosen order and may define an effective time-independent Hamiltonian for a given time, T , yielding $\hat{H}_{\text{eff}} = \frac{i\hbar}{T} A(T)$. We can immediately make two simplifying assumptions: first concerns the fast oscillating A_1 term. This looks like separate spin dependent phase-space loops originating from the off-resonant SDFs. As these loops close with periodicity $\tau = 2\pi/\Delta$, we may select a gate duration T such that $T = n\tau$, to enforce

$A_1(T) = 0$. Second, as with the geometric phase gates seen in section ??**phase gate section**, the second order Magnus term contains self-terms that appear as the accumulation of a geometric phase. In the case that this geometric phase acts globally on the system, it can be ignored. If the accumulated phase is spin, or oscillator state dependent, then interesting dynamics may be observed, however these effects are out of the scope of this thesis. With these two assumptions, we may now consider how to recover the general family of non-linear bosonic interactions given in table 1.1

We set $\phi_\alpha = 0$, $\phi_\beta = \phi$, $\sigma_\alpha = \sigma_x$, $\sigma_\beta = \sigma_y$, and $\delta_\beta = m\delta_\alpha$, where m is some integer. Integer m also helps to select durations that both excursions of A_1 close. For $m = +1$, we find that the effective Hamiltonian is,

$$H_{\text{eff}} = \dots \tag{A.32}$$

This narrative is a bit tangled. I think potentially there is a simpler way to get to the effective interactions...

References

- [1] O. Băzăvan et al. *Squeezing, trisqueezing, and quadsqueezing in a spin-oscillator system*. en. arXiv:2403.05471 [physics, physics:quant-ph]. 2024.
- [2] S. Saner et al. *Generating arbitrary superpositions of nonclassical quantum harmonic oscillator states*. arXiv:2409.03482 [quant-ph]. 2024.
- [3] Jordan Cotler et al. “Quantum Virtual Cooling”. In: *Phys. Rev. X* 9.3 (2019). arXiv:1812.02175 [quant-ph].
- [4] Bálint Koczor. “Exponential Error Suppression for Near-Term Quantum Devices”. In: *Phys. Rev. X* 11.3 (2021).
- [5] William J. Huggins et al. “Virtual Distillation for Quantum Error Mitigation”. In: *Phys. Rev. X* 11.4 (2021).
- [6] Yuan Liu et al. “Hybrid Oscillator-Qubit Quantum Processors: Instruction Set Architectures, Abstract Machine Models, and Applications”. In: *PRX Quantum* 7.1 (2026).
- [7] Lukas Brenner et al. *Factoring an integer with three oscillators and a qubit*. en. arXiv:2412.13164 [quant-ph]. 2024.
- [8] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. en. ISBN: 9780511976667. 2010.
- [9] Seth Lloyd and Samuel L. Braunstein. “Quantum Computation over Continuous Variables”. In: *Phys. Rev. Lett.* 82.8 (1999).
- [10] Axel M. Eriksson et al. “Universal control of a bosonic mode via drive-activated native cubic interactions”. en. In: *Nat Commun* 15.1 (2024).
- [11] R. T. Sutherland and R. Srinivas. “Universal hybrid quantum computing in trapped ions”. en. In: *Phys. Rev. A* 104.3 (2021). arXiv:2105.05768 [physics, physics:quant-ph].
- [12] S. Saner et al. *Real-Time Observation of Aharonov-Bohm Interference in a $\mathbb{Z}_2$ Lattice Gauge Theory on a Hybrid Qubit-Oscillator Quantum Computer*. en. arXiv:2507.19588 [quant-ph]. 2025.
- [13] Oana Bazavan. “Easy as XYZ: Exploring non-commuting coherent interactions in trapped ions”. en. In: ().
- [14] Sebastian Saner. “Quantum Control of Hybrid Spin-Oscillator Systems”. en. In: ().
- [15] Christa Flühmann and Jonathan P. Home. “Direct characteristic-function tomography of quantum states of the trapped-ion motional oscillator”. en. In: *Phys. Rev. Lett.* 125.4 (2020). arXiv:1907.06478 [physics, physics:quant-ph].
- [16] V. M. Schäfer et al. “Fast quantum logic gates with trapped-ion qubits”. en. In: *Nature* 555.7694 (2018). Number: 7694.

- [17] Zhenyu Cai et al. “Quantum error mitigation”. In: *Rev. Mod. Phys.* 95.4 (2023).
- [18] M. D. Bowdrey et al. “Fidelity of Single Qubit Maps”. en. In: *Physics Letters A* 294.5-6 (2002). arXiv:quant-ph/0201106.
- [19] Michael A Nielsen. “A simple formula for the average gate fidelity of a quantum dynamical operation”. In: *Physics Letters A* 303.4 (2002).
- [20] Jonas Helsen et al. “A new class of efficient randomized benchmarking protocols”. en. In: *npj Quantum Inf* 5.1 (2019).
- [21] Shelby Kimmel et al. “Robust Extraction of Tomographic Information via Randomized Benchmarking”. en. In: *Phys. Rev. X* 4.1 (2014).